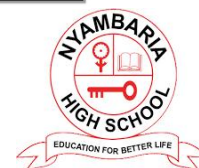
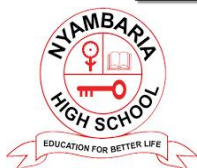


NYAMBARIA BOYS' 2025 PREDICTION



KCSE PROJECTION EXAM

PHYSICS

232/1

PAPER 1 (THEORY)

TIME: 2 HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education

INSTRUCTIONS TO THE CANDIDATE:

- (a) Write your **name** and **index number** in the spaces provided above.
- (b) **Sign** and write the **date** of examination in the spaces provided above.
- (c) This paper consists of **two** Sections **A** and **B**.
- (d) Answer **all** the questions in sections **A** and **B** in the spaces provided.
- (e) All working **must** be clearly shown in the spaces provided.
- (f) Mathematical tables and electronic calculators **may be** used.

FOR EXAMINER'S USE ONLY:

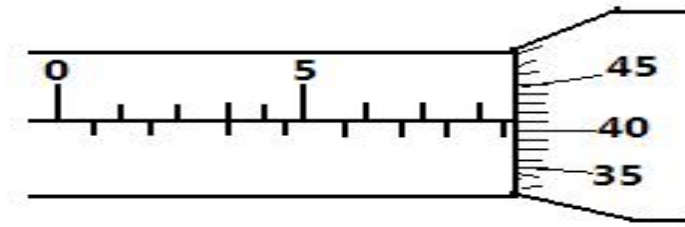
Section	Question	Maximum Score	Candidate's Score
A	1 – 13	25	
	14	11	
	15	10	
B	16	13	
	17	12	
	18	9	
Total Score		80	

SECTION A (25 MARKS)

(Answer ALL the questions in the spaces provided)

1. What is the reading on the micrometer screw gauge shown below with an error of +0.5mm?

(1mk)



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2. In the study of free fall, it is assumed that the force F acting on a given body of mass, m , is gravitational, given by $F = ma$. State **two** other forces that act on the same body (1mk)

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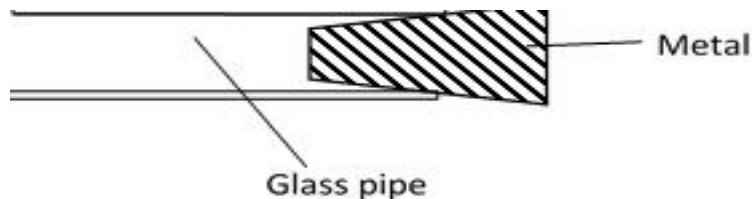
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3. State **two** facts which show that heat from the sun does not reach the earth surface by convection. (2mks)

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4. The figure below shows a piece of metal stuck in a hollow glass pipe. Explain how temperature change may be used to separate them (2mks)



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5. The water in a burette is 30.6cm^3 , 50 drops of water each of volume V are added to the water in the burette. The final reading of the burette was 20.6cm^3 . Calculate the radius of the drop of water.

(2mks)

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6. In an experiment to demonstrate Brownian motion, smoke was placed in a smoke cell and observed under a microscope. State and explain the observation.

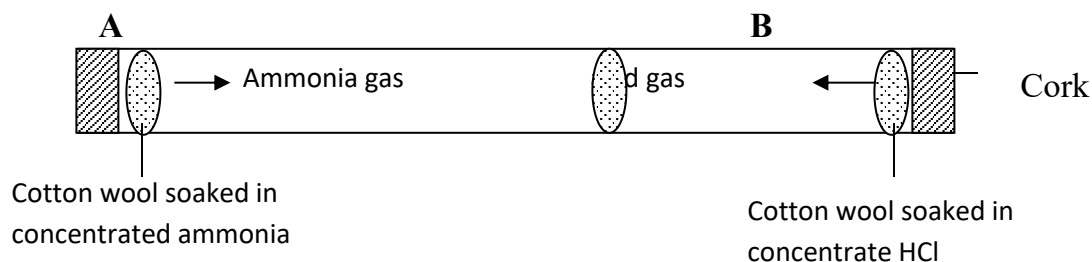
(2mks)

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7. In the figure below, ammonia gas and an acid gas diffuse and react to form a white deposit on the walls of a long glass tube as shown.



- (i) What conclusion can be made from the result of this experiment?

(1mk)

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- (ii) How does the size and mass of a gas affect its rate of diffusion?

(1mk)

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- (iii) The experiment is performed at a lower temperature. Explain how the time taken to form the white deposit would be affected.

(1mk)

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8. Determine the change in momentum produced when a force of $3500N$ acts on a body which is at rest for 0.002 seconds. (2mks)

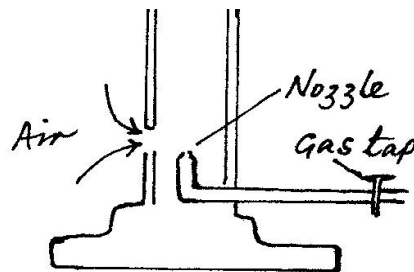
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9. The figure below shows a Bunsen burner. Explain how air is drawn into the burner when the gas tap is opened.

(2mks)



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10. A mass of $100g$ is hung at the $10cm$ mark and a $50g$ mass at the $70cm$ mark of a uniform metre rule balanced at the $40cm$ mark. Determine the weight of the rule. (3 marks)

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11. Four students were playing football during which the ball got deflated. Explain what happened to its density

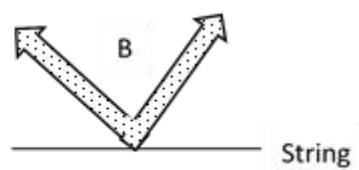
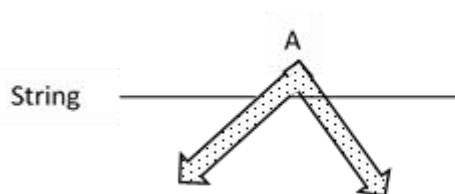
(2mks)

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12. State the branch of physics that deal with kinetic energy of matter. (1mk)

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13. A student balances a V - shaped uniform wire on a tight string as shown in figure A and B. state with reason the one which is easier to do (2mks)



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SECTION B (55 MARKS)

14. A student in LubinuGirls set up an experiment to study the acceleration of a trolley using ticker tape timer. The timer made 50 dots per second on the tape. Dots A to E measured 2.5cm apart and dots E to I measured 4.5cm apart.

a) Using a scale drawing show the dots A, B C, D, E, F, G and I as they appeared on the tape. (3mks)

b) Determine the velocity of the trolley from:

i) A to E. (2mks)

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ii) E to I. (2mks)

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c) Calculate the acceleration of the trolley. (2mks)

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d) What end of the tape was fixed onto the trolley? (1mk)

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e) State **two** precautions that the student should take before she takes her final samples of the dots. (2mks)

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15. a) An astronaut in orbit round the earth may feel weightless even when the earth's gravitational field still acts on him. **Explain**
(2mks)

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b) Distinguish between angular velocity and linear velocity
(1mk)

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c) A stone is whirled with uniform speed in a horizontal circle of radius 15 cm. it takes the stone 10 seconds to describe an arc of length 4cm. calculate

i) Angular velocity
(3mks)

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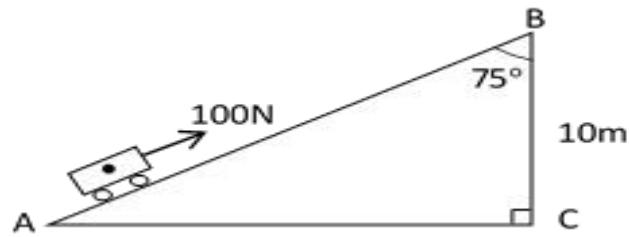
ii) Linear velocity of the stone
(2mks)

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iii) Periodic time
(2mks)

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16. The figure **below** shows an inclined plane on which a trolley of mass 30kg is pulled up a slope by a force of 100N, parallel to the slope. The trolley moves so that its centre of mass travels from points A to B.



(i)Determined the work done on the trolley against the gravitational force in moving from **A** to **B**.

(2mks)

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(ii) Determine the work done by the force in moving the trolley from A to B. (3mks)

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(iii) Determine the percentage of the work input that goes to waste
(3mks)

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(iv) Determine the frictional force.
(1mk)

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(v) Determine the mechanical advantage of the system.
(1mk)

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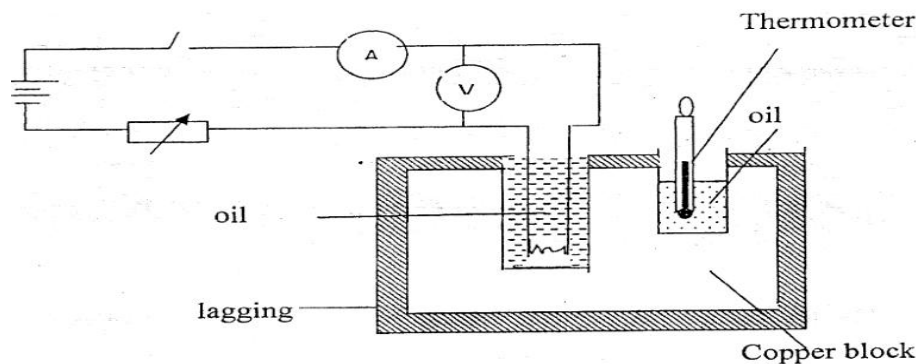
(vi) Find the velocity ratio
(1mk)

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17. a. The figure below shows a set-up that can be used to determine the specific heat capacity of a metal block.



I) Other than temperature and current, state **two** measurements that should be taken in the experiment to determine the specific heat capacity of the block. (2mks)

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II) Describe how the method can be used to determine the specific heat capacity of the metal block.

(3mks)

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III) State the purpose of oil in the set-up.

(1mk)

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(b) A well lagged copper can together with a copper stirrer of total heat capacity 60JK^{-1} contains 200g of water at 20°C . Dry steam at 100°C is passed in while the water is stirred until the content reach a temperature of 50°C . Determine the mass of condensed steam. (Specific latent heat of vaporization of water is $2.26 \times 10^6 \text{ J/kg}$ and specific heat capacity of water is 4200 J/kgK) **(4mks)**

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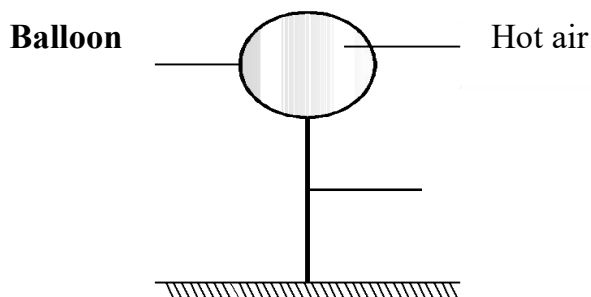
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18. a) State the law of floatation
(1mk)

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b) The diagram below shows a hot air balloon tethered to the ground on a calm day. The balloon contains 1300m^3 of hot air of density 0.82 kg/m^3 . The mass of the material making the balloon without hot air is 420kg . The density of the surrounding air is 1.35 kg/m^3 . Determine



String

i) The total weight of hot air **balloon**

(3mks)

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ii) The weight of air displaced by the **balloon**

(2mks)

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iii) Upthrust force on the balloon

(1mk)

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iv) The tension in the rope holding the balloon in the ground.

(2mks)

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i) The acceleration with which the balloon begins to rise when released.

(3mks)

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